

Question	Answer	Marks
13(a)(i)	probability of decay (of a nucleus)	M1
	per unit time	A1
13(a)(ii)	$A = A_0 e^{-\lambda t}$ after one half-life, $\frac{1}{2}A_0 = A_0 e^{-\lambda t_{1/2}}$	M1
	$\frac{1}{2} = \exp(-\lambda t_{1/2})$ and hence taking logs, $\ln 2 = \lambda t_{1/2}$	A1
13(b)	activity = $3.8 \times 10^4 \exp(-\ln 2 \times 36 / 15)$	C1
	= 7200 Bq	C1
	or activity = $3.8 \times 10^4 / 2^{2.4}$	(C1)
	= 7200 Bq	(C1)
	volume = $(7200 / 1.2) \times 5.0$	C1
	= $3.0 \times 10^4 \text{ cm}^3$	A1
	OR activity of 5.0 cm^3 = $1.2 \times 2^{2.4}$	(C1)
	= 6.3336 Bq	(C1)
	volume = $(3.8 \times 10^4 / 6.3336) \times 5.0$	(C1)
	= $3.0 \times 10^4 \text{ cm}^3$	(A1)